

Circumgalactic dust halos update #2

Jacqueline McCleary

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1 Executive Summary

This new report is organized as follows:

1. Literature review (§2): timely (like, from 2023-2024) advances in modeling and theory have added new relevance to our results. These include a new extinction law, a new model for dust self-shielding motivated by a need to explain the Ménard result, and an improved galactic extinction model.
2. A description of modeling updates and code improvements.
3. Updated results based on new models, including the fitting of power laws! The TL;DR is that while the *slope* for all redMaGiC-based fits is a close match to the Ménard result, the *amplitude* is about twice as high. This is something we can and should look into in future work.
4. For the first time, I have also expand our search for extended dust halos to the Northern Sky! I tried a few different SDSS background samples as well as two different foreground samples: WISExSCOS and GWSLC (itself divided by specific star formation rate). foreground samples. While the WISExSCOS x SDSS results tended to be consistent with the redMaGiC samples at *low* impact parameters, at *high* impact parameters ($> XX$ arcminutes), the extinction flattens out and doesn't decrease. Meanwhile, the GWSLC x SDSS extinction has an interesting dip at intermediate impact parameters but agrees with the redMaGiC curves at high and low impact parameters.
5. Also, my talk at the Aspen Center for Physics dust workshop went really well! There were good questions and a lot of interest afterwards.

In general, I have tried to keep this kind of brief as an exercise for synthesizing the intense work of the last year as a prelude to writing up the paper.

2 Literature Review

Since my last update over a year ago, several exciting and highly germane papers have been published. Results include an interesting hydrosim study of dust survival in the IGM, a new model for dust extinction, and an update to the canonical SFD map.

2.1 Gordon et al. 2023

My previous results were based on dust models published around 2000, like Calzetti et al. 2000 and Fitzpatrick 1999. When I presented these results at the Aspen Center for Physics workshop, Daniella Calzetti told me that she was flattered that I was using her extinction law but that it was out of date. She pointed me to an updated model published by Karl Gordon et al. 2023, hereafter G23 [2]). Using four independent spectroscopically measured dust extinction curves and dozens of individual sightlines, G23 presents a single R_V -dependent extinction relation from 912 Å to 32 microns. In their words, the main advance of their approach is that “all previous $R(V)$ relationships were based on a combination of spectroscopic and photometric measurements and/or did not cover the full wavelength range.” Their principal result, the A_λ/A_V relation, is shown in Figure 8 of that paper. Their extinction relation, shown in Figure 8 of that paper, is obtained by first fitting a linear A_λ/A_V at each wavelength; polynomials are then fit to the resulting coefficients piecewise in five wavelength regimes (FUV, NUV, optical, near- and mid-IR); the polynomials are smoothed at the boundaries to obtain a continuous function (see Equation 5 of that paper). They find that the characteristic FUV rise and 2175 Å bump are dependent on R_V , but that 10 and 20 micron features are not. The largest offsets between their extinction laws and other, comparable studies are in the FUV and the NIR (cf. Figure 8, lower panels). Interestingly, their extinction relation lies in the middle of the older studies’ extinction relations. In other words, the G23 result appears to be a “golden mean.”

2.2 Helena outflow paper

At the Aspen dust conference, I reconnected with an old friend from the UC Santa Cruz High Performance Astrophysical Computing Center (HiPACC) summer school in 2010. He is now a professor at UVA, and his graduate student Helena Richie gave a presentation of hydrodynamical simulations of dust evolution in galactic outflows, which is described in [2024ApJ...974...81R]. Specifically, they use the Cholla code to set up “wind tunnel” simulations of cool, dusty clouds driven into the hot halo by star formation activity. They find that that 60% of 0.1-micron sized dust grains entrained in these galactic outflows could survive to populate both the hot and cool phases of the halo. Meanwhile, grains smaller than 0.01 microns cannot withstand hot-phase exposure and must be created in situ.

2.3 CSFD

I saw Chiang give a talk on this topic at the Aspen dust workshop, and then my COSMOS-Web colleague Nao Suzuki called my attention to this result again at the Tokyo meeting. The basic premise of the paper is that the classic Schlegel, Finkbeiner, & Davis (1998, SFD) dust map has significant contamination from the cosmic infrared background, that is, the two-halo term of large-scale structure. Apparently, this fact has been discovered and rediscovered in the literature [10.1093/pasj/59.1.205, 10.1093/pasj/65.2.43], including by our friend Brice Ménard [2019ApJ...870..120C].

2.4 Dust as a supernova systematic

Additionally, intergalactic dust introduces significant scatter to Type 1a supernova (SN 1a) distance estimates. According to [2023arXiv231112098R], dust is the second-largest source of uncertainty in SNe Type 1a cosmological parameter constraints, accounting for 11.5% of the total variance. Current SNe analyses include intergalactic dust reddening as a function of wavelength and SN

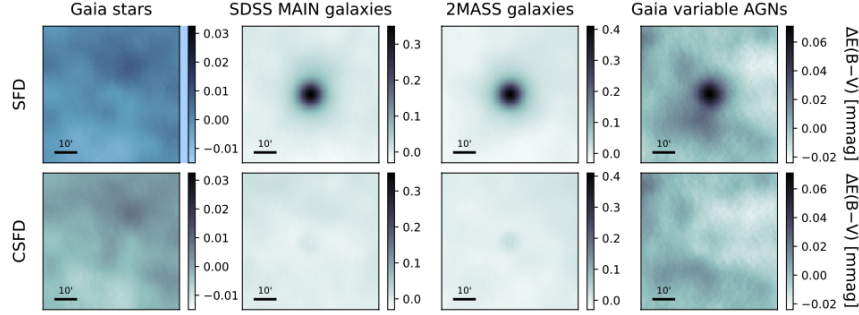


Figure 1: Stacked images of the SFD $E(B - V)$ (top row) and the CSFD equivalent (bottom row) centered on different types of sources. From left to right, the stacks are centered on Gaia stars, SDSS main sample galaxies, 2MASS galaxies, and Gaia AGNs, respectively. Aside from the stars, the SFD $E(B - V)$ values around all galaxy samples are biased high due to the LSS contamination. This bias is significantly reduced in CSFD. (Caption and figure adapted from [2023ApJ...958..118C].)

redshift in the systematic error budget, but because the magnitude of the effect is somewhat uncertain, do not correct the distance measures themselves. My COSMOS-Web colleague, Nao Suzuki, pointed out that if we can differentiate extinction laws according to foreground galaxy properties, as we appear to do, then we could derive a more precise calibration for foreground reddening of supernovae.

3 Code and Modeling Updates

3.1 Code updates

3.1.1 Color-matched randoms

- describe algorithm, show nice figure
- This didn't significantly affect the result, however.

3.1.2 Random catalog generation overhaul

- Mention bug of picking healpixels; could even show the bad ra/good ra figures
- Also implemented ability to generate catalogs in galactic coordinates, which made sense for the WISExSCOS catalog
- Based on the redMaGiC ratio of random galaxies to real galaxies, the random catalogs I create are $\sim 20\times$ the length of their respective data catalogs. For very large galaxy catalogs like SDSS, generating an equivalent random catalog with almost a billion objects is too much for my HPC to handle. (Or, perhaps my code is inefficient.) So, I added a little option to create multiple smaller catalogs and concatenated them.

3.1.3 Treecorr variance and misc. utilities

- Fixed variance! Used patches and compensated estimates
- RADec_to_healpixel

- dereddened magnitudes, the script used to incorporate CSFD extinction values into magnitudes.

3.2 Model improvements

3.2.1 Gordon 2023 extinction

The G23 extinction model is available through the `dust_extinction` software package [1].¹ Implementing it in the code was very simple and amounted to adding an import statement for this newer `dust_extinction` package in the `src/extinction_model.py`. Both G23 and the older CCM89 models are available in `dust_extinction`; the `extinction` Python package is still imported in case someone wants to see results with other old models: Calzetti00, odonnell94, fitz99. As before, the default R_V is 3.1 but any reasonable value is available.

3.2.2 CSFD

- As mentioned earlier, Nao pointed me to this CSFD paper [2023ApJ...958..118C]. Since the amplitude of the residual seemed non-negligible (10^{-4}) it was worth implementing.
- I located a nice software package called `dustmapper`² to query the CSFD dust map and obtain extinction coefficients for a given set of coordinates. Specifically, there is a handy `dustmaps.csfd.CSFDQuery` method that takes as input an RA, Dec list in SkyCoords format and returns an array of E(B-V) values. An R_V value is used to convert those into extinction coefficients and then used to de-redden SDSS and DES magnitudes. Using SFD maps to obtain E(B-V), then apply band-specific extinction coefficients to get extinction values.
- Subtracting these extinction values $A_{X,CSFD}$ from the observed magnitudes MAG_X to obtain the dereddened magnitudes:

$$MAG_X_CORRECTED = MAG_X - A_{X,CSFD}$$

For the DES magnitudes, I used the corrected magnitude prescription of Pandey et al. 2021:

$$MAG_CORRECTED = MAG + DELTA_MAG_Y4 + DELTA_MAG_CHROM - A_{X,CSFD}$$

- (placeholder for pseudocode of algorithm.)
- As one might expect, this had a notable effect on the amplitude of the signal, increasing it by XX%.

3.2.3 Weighted Least-squares fitting.

To fit models to our results and facilitate comparison to the literature, I have added a function (`do_WLS_fit.py`) to perform weighted least-squares fits to extinction profiles. Surprisingly, we hadn't implemented this yet! The function, partially reproduced in Appendix A, is principally a wrapper for the StatsModel package's weighted least-squares method, with the additional capability to rescale weights for fits in logarithmic space (i.e., power laws). This code was used to generate all fits shown in this report, and weights are based on TreeCorr's estimates of the variance at each point in the extinction profile.

¹https://github.com/karllark/dust_extinction

²<https://github.com/greggreen/dustmaps>

4 Catalogs: Round 2

4.1 redMaGiC and WISExSCOS

- I have a note in here about going over magnitude column definitions, but I don't know what that means. Perhaps just the names?
- List counts for randoms and reals
- The WISExSuperCOSMOS catalog is trimmed to $0 < z < 0.2$, rather than $z < 0.11$ as before, as well as $17 < bCalCorr < 21$ (really with that last one?)
- To ensure a decent redshift separation between the background and foreground samples, the high-density redMaGiC sample is restricted to $z > 0.5$ (the full redshift range is $0.15 < z < 0.65?$).
- There is still no WISExSuperCOSMOS random catalog available, so one is created using the corrected randoms algorithm described in Section 3.1 in galactic coordinates to match the original catalog coordinates.

4.2 SDSS and GWSLC

- Detail catalog counts, selections, etc. including magnitude columns.
- Show SDSS query, database
- GWSLC cuts: used X-2 variant of available catalogs, which combines the deepest of the all-sky, medium, and deep surveys. I had originally hoped to examine dust as a function of galaxy type and that is the recommended catalog.
- Don't forget to tabulate wavelengths!

4.3 Systematics testing

- I am always worried that I am screwing up somehow so to test for systematic error I perform my analysis using star catalogs.
- Stars should have no dust so they make good test catalogs either in foreground or in background.
- Randoms: there are strong gradients in density across the fields (makes sense, I guess) which might (seemed to?) affect the results.
- So, to create random catalogs with a matching density gradient, I grabbed larger samples of stars to serve as randoms.

4.3.1 Gaia stars

- This was used as a foreground sample for the redMaGiC catalogs, so it got used for correlations twice: once for the high-density sample, once for the high-luminosity high-redshift sample.
- I had an oddly tough time narrowing down the survey area on Cosmohub, so I ended up grabbing a fairly large area over the Southern sky and applying the redMaGiC mask. Final count: 478,494 objects.

- Detail catalog cuts, which I don't even remember!
- Show one of my pretty overlap plots!

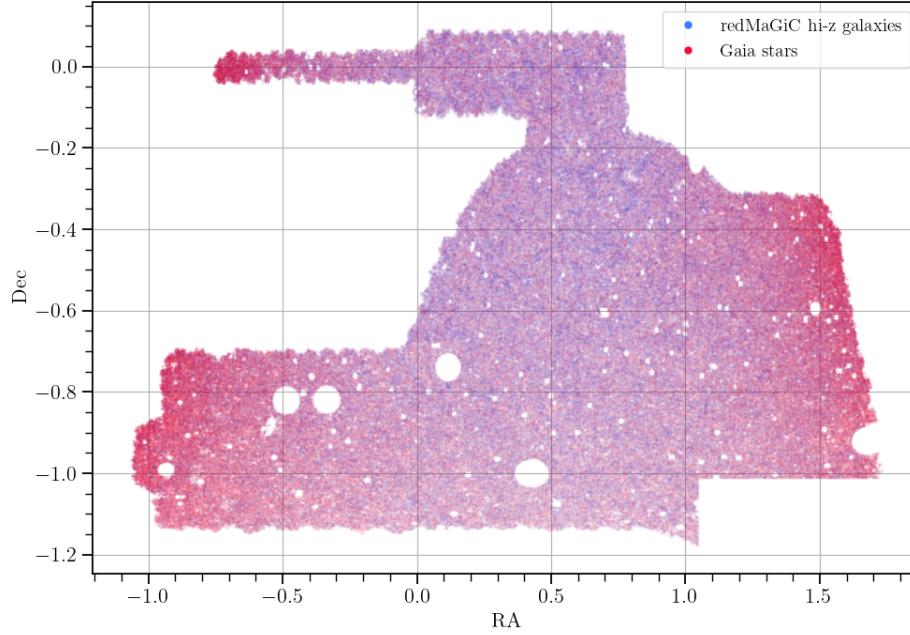


Figure 2: default

4.3.2 DES Y3 stars

- This was used as a background sample, standing in for the redMaGiC catalogs in a correlation with WISExSuperCOSMOS
- Show catalog query used, which was taken from [\[2021ApJS..254...24S\]](#).
- Detail catalog cuts made, which include $18 < \text{mag}_r\text{-corr} < 21$.
- After masking and the like, there were 824,345 stars in the “real” catalog and 9,892,452 randoms.

5 Results

- Comments for the individual samples are below, but globally, both the SFD and CSFD results agrees with the *slope* of the [\[2010MNRAS.406.1815M\]](#) power law (Equation 13 in that reference) within one sigma. However, the *amplitude* of the CSFD relationship is twice as high as the [\[2010MNRAS.406.1815M\]](#) result. I am not entirely sure what to make of this, but it is what it is.
- Could discuss the SFD results...

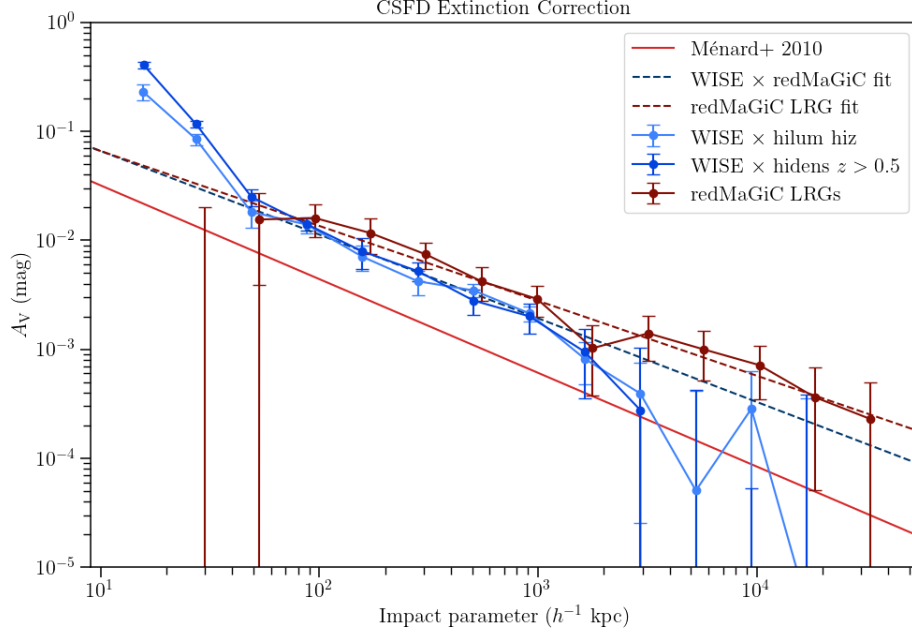


Figure 3: Extinction vs impact parameter for WISExSuperCOSMOS (blue points) and redMaGiC LRGs (maroon points). Results are based on the CSFD extinction correction.

5.1 RedMaGiC-based results

5.1.1 WISExSuperCOSMOS foreground

The extinction profiles of WISExSuperCOSMOS galaxies as measured with redMaGiC background galaxies are shown in Figure 3: the result from cross-correlation with the $z > 0.5$ high-density redMaGiC sample is shown in dark blue; the result from cross-correlation with the high-redshift redMaGiC sample is shown in light blue. In both cases, the extinction profiles follow a broken power law, with a rapid decrease from $A_V \sim 0.3$ to $A_V \sim 0.02$ in the inner $50 \text{ kpc } h^{-1}$, then a slower decline to $A_V \sim 0.02$ at $1 \text{ Mpc } h^{-1}$. At the transition to the 2-halo regime ($r \gtrsim 2 \text{ Mpc } h^{-1}$), the extinction profiles rapidly diminish to below our detection threshold ($A_V \sim 10^{-5}$).

I fit power laws to the WISExSuperCOSMOS extinction profiles within the 1-halo regime ($50 \text{ kpc } h^{-1} < r < 1 \text{ Mpc } h^{-1}$); the inner limit is set to the point where the redMaGiC LRG extinction is non-zero and the outer limit to the points where the extinction begins to roll off.

The best-fit power law to the extinction profile obtained with the $z > 0.5$ high-density redMaGiC sample is given by Equation 1:

$$A_V(r) = (1.14 \pm 0.08) \times 10^{-2} \left(\frac{r}{h^{-1} \text{ kpc}} \right)^{-0.84 \pm 0.04} \quad (1)$$

Similarly, for the extinction profile obtained with the $z > 0.5$ high-lum high- z redMaGiC sample is given by Equation 2:

$$A_V(r) = (1.02 \pm 0.11) \times 10^{-2} \left(\frac{r}{h^{-1} \text{ kpc}} \right)^{-0.72 \pm 0.06} \quad (2)$$

Combining the two WISExSuperCOSMOS $\times$ redMaGiC profiles, the best fit power law is Equation 3; this is the result I plan to publish.

$$A_V(r) = (1.08 \pm 0.07) \times 10^{-2} \left(\frac{r}{h^{-1} \text{ kpc}} \right)^{-0.77 \pm 0.04} \quad (3)$$

5.1.2 redMaGiC LRG

The extinction profile of the LRG galaxy sample ($z < 0.45$ high-density redMaGiC galaxies cross-correlated with the high-luminosity high-redshift redMaGiC catalog) is shown in maroon in Figure 3. As might be expected for this foreground population, there is little extinction at the lowest impact parameters ($r < 50 \text{ kpc}$). While qualitatively similar to the SuperCOSMOS-based results in the one-halo regime, the LRG sample shows significant extinction well into the two-halo regime. The best-fit power law to this profile is given by Equation 4:

$$A_V(r) = (0.82 \pm 0.19) \times 10^{-2} \left(\frac{r}{h^{-1} \text{ kpc}} \right)^{-0.69 \pm 0.06} \quad (4)$$

The slope and amplitude of Equation 4 agree with the combined redMaGiC fit of Equation ?? to within about 1σ through the one-halo regime ($r \lesssim 1 \text{ Mpc}$).

5.1.3 Comparison to SFD

Given the acknowledged problems of SFD98, the CSFD formulation is the default Milky Way extinction correction. It may still be of interest to compare the profiles obtained with the SFD and CSFD prescriptions; this is shown for the three redMaGiC-based profiles in Figure 4. I noticed a couple of interesting trends here. First, there didn't seem to be much of an effect for the high-density background catalog, but using the SFD extinction correction, the high-lum high- z sample has more extinction at high impact parameters. Put differently, it appears that the SFD prescription added too much extinction at high impact parameters, and with CSFD the two redMaGiC catalogs produce consistent results for WISExSuperCOSMOS. On the other hand, compared to the SFD98 prescription, the CSFD prescription added a significant amount of dust at higher impact parameters for the redMaGiC LRG sample, leading to significant reddening past $1 \text{ Mpc } h^{-1}$. Although quite different from the effect for the WISExSuperCOSMOS result, this behavior for LRG is consistent with the idea that SFD98 overcorrected foreground extinction due to contamination from the LSS. If one assumes that the LRG is a particularly biased tracer of the LSS_i, then it makes sense that the CSFD prescription adds some reddening back in.

For the sake of comparison, I also fit power laws to extinction profiles resulting from the SFD98 Milky Way foreground prescription. The best-fit power law to the extinction profile obtained with the $z > 0.5$ high-density redMaGiC sample is given by Equation 5:

$$A_V(r) = (5.9 \pm 2.3) \times 10^{-3} \left(\frac{r}{h^{-1} \text{ kpc}} \right)^{-0.62 \pm 0.12} \quad (5)$$

The best-fit power law to the extinction profile obtained with the high-lum high- z redMaGiC sample is given by Equation 6:

$$A_V(r) = (9.5 \pm 0.9) \times 10^{-3} \left(\frac{r}{h^{-1} \text{ kpc}} \right)^{-0.75 \pm 0.03} \quad (6)$$

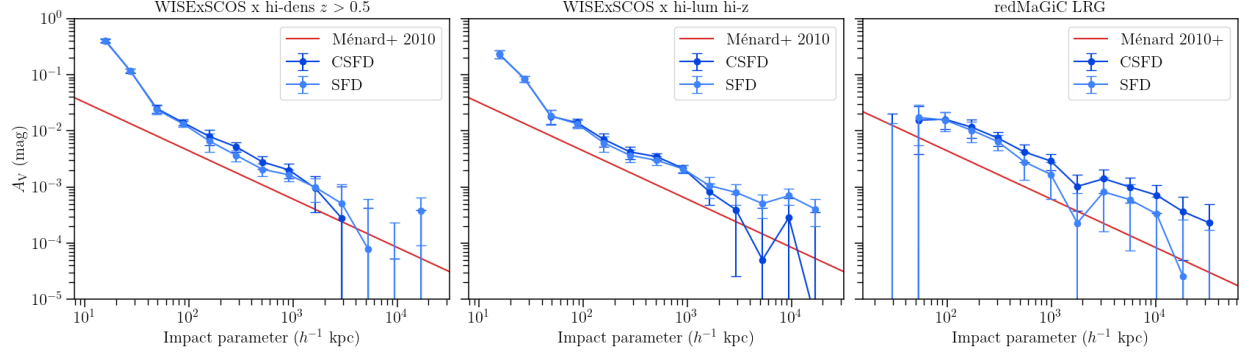


Figure 4: Extinction vs impact parameter for redMaGiC-involved fits. CSFD was used in all cases.

The result of a fit to the combined WISExSuperCOSMOS x redMaGiC samples is

$$A_V(r) = (8.7 \pm 1.3) \times 10^{-3} \left(\frac{r}{h^{-1} \text{ kpc}} \right)^{-0.74 \pm 0.08} \quad (7)$$

As above, Equations 5–7 describe the flatter parts of the profiles ($50 \text{ kpc } h^{-1} < r < 1 \text{ Mpc } h^{-1}$). Finally, the LRG profile is best fit by Equation 8

$$A_V(r) = (6.9 \pm 2.1) \times 10^{-3} \left(\frac{r}{h^{-1} \text{ kpc}} \right)^{-0.87 \pm 0.14} \quad (8)$$

For this last profile, my WLS algorithm just would not produce a reasonable fit, which forced me to adopt a conventional least-squares algorithm for the LRG fit. The amplitude of Equation 8 matches the fit of Equation 4 within error bars, however, the slopes are in a 3σ disagreement. Meanwhile, the fits of Equations 5–7 agree with 1–3 within 1σ .

5.2 Systematics testing results

This section summarizes the results of the various systematics tests that I implemented using star catalogs. In general, these are mostly consistent with zero, though with noticeable residuals. Then again, these are 10^{-4} results. All systematics testing results are based on the CSFD extinction corrections.

5.2.1 Gaia foreground x redMaGiC background

- Show the fit results for hi-lum and separately
- Can discuss the fact that high-density sample has a small residual, though I am not entirely sure why

$$A_V(r) = (3.1 \pm 4.2) \times 10^{-4} \left(\frac{r}{h^{-1} \text{ kpc}} \right)^{-6.7 \times 10^{-5} \pm 8.9 \times 10^{-5}} \quad (9)$$

$$A_V(r) = (-3.7 \pm 3.6) \times 10^{-5} \left(\frac{r}{h^{-1} \text{ kpc}} \right)^{1.1 \times 10^{-5} \pm 7.6 \times 10^{-5}} \quad (10)$$

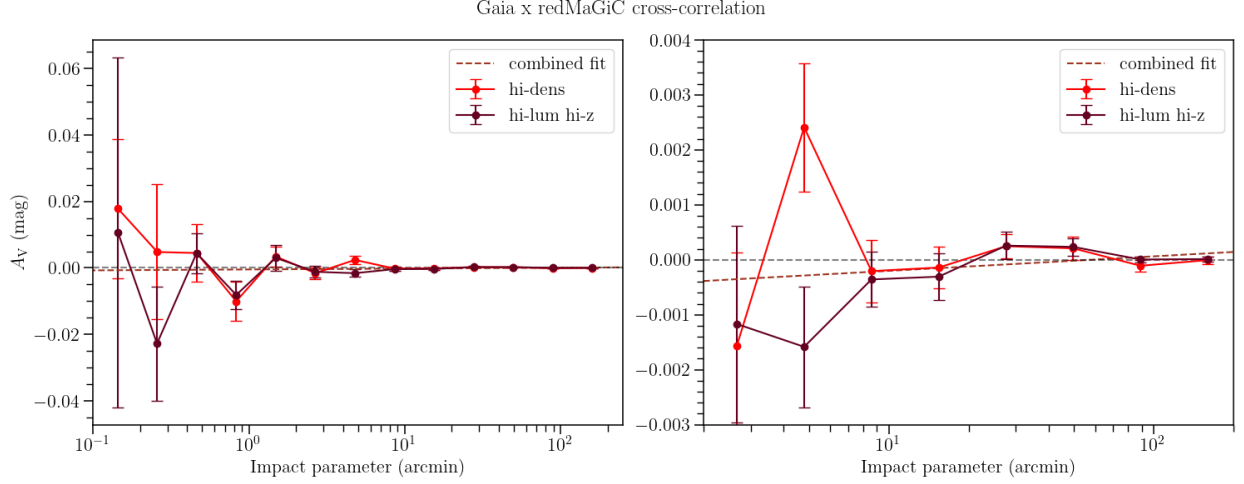


Figure 5: Gaia star foreground catalog cross-correlated with redMaGiC high-luminosity high-redshift (maroon) and high-density (red) background catalogs. The left panel shows the full impact parameter range; the right panel is cropped to show values at higher impact parameters.

$$A_V(r) = (1.1 \pm 2.6) \times 10^{-4} \left(\frac{r}{h^{-1} \text{ kpc}} \right)^{-2.1 \times 10^{-5} \pm 5.6 \times 10^{-5}} \quad (11)$$

$$A_V(r) = (-4.1 \pm 1.8) \times 10^{-4} \left(\frac{r}{h^{-1} \text{ kpc}} \right)^{1.4 \times 10^{-4} \pm 6.4 \times 10^{-4}} \quad (12)$$

$$A_V(r) = (-5.2 \pm 1.5) \times 10^{-4} \left(\frac{r}{h^{-1} \text{ kpc}} \right)^{9.4 \times 10^{-5} \pm 5.4 \times 10^{-4}} \quad (13)$$

$$A_V(r) = (-4.7 \pm 1.1) \times 10^{-4} \left(\frac{r}{h^{-1} \text{ kpc}} \right)^{1.1 \times 10^{-4} \pm 4.0 \times 10^{-4}} \quad (14)$$

Excluding the first two points, which seemed to skew the fits, the results of fitting are shown below. I decided to go with least-squares fitting because I liked the look of it better.

5.2.2 SuperCOSMOSxWISE x DES Y3 stars

- Show figures and fit results for this sample
- Again, small but mildly significant residual.

I decided to go with the least-squares, because the weighted least squares looked like nonsense for this fit. Much bigger slope than made any sense.

$$A_V(r) = (-5.3 \pm 4.5) \times 10^{-4} \left(\frac{r}{h^{-1} \text{ kpc}} \right)^{1.8 \times 10^{-4} \pm 1.3 \times 10^{-4}} \quad (15)$$

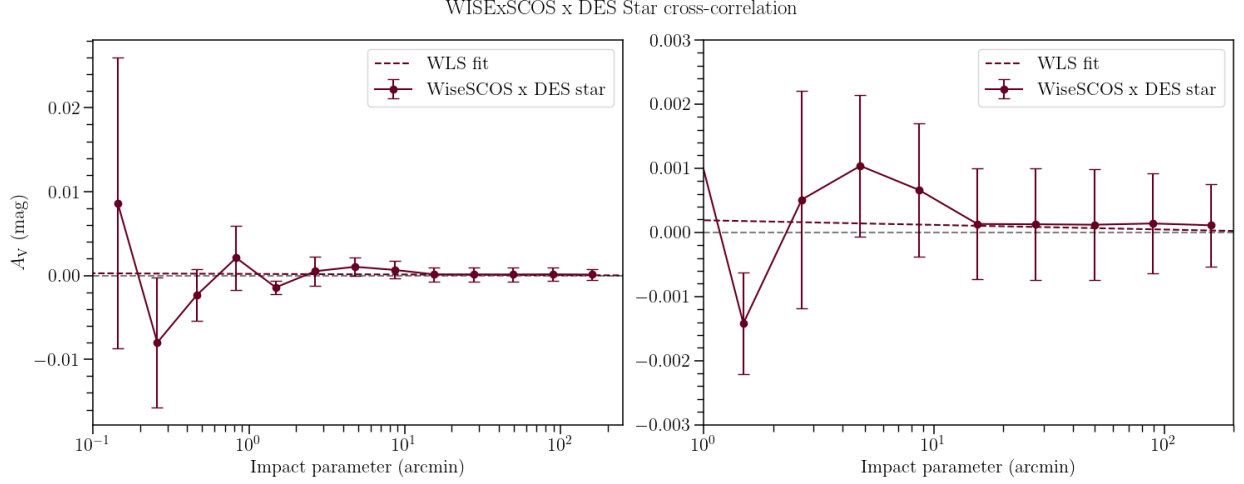


Figure 6: WISExSuperCOSMOS foreground cross-correlated with stars culled from the DES Y3 catalog. The left panel shows the full range of impact parameters considered; the right panel is cropped to show values at higher impact parameters only.

$$A_V(r) = (1.9 \pm 1.3) \times 10^{-4} \left(\frac{r}{h^{-1} \text{ kpc}} \right)^{-3.2 \times 10^{-5} \pm 4.7 \times 10^{-4}} \quad (16)$$

5.3 SDSS-based results

These results are based on two separate foreground catalogs: the same WISExSuperCOSMOS catalog used for the southern sky, DES-based catalogs, and the GWSLC catalog described above. I will go through these results separately. These are certainly interesting to see because so far we hadn't extended the search to the Northern sky. However, I don't consider the results publication-ready.

That said, may be an argument to be made for publishing only the *low* impact parameter points. given Eric's comments about the significance of a detection of a lot of dust at significance, They consistently line up quite well with the redMaGiC-based results at these impact parameters, and Eric did note the significance of the detection of extended dust halos in what is really the extended CGM.

5.3.1 WISExSuperCOSMOS x SDSS

- On the one hand, it's cool to see this, especially since this is the same foreground sample we use elsewhere that gave us successful results.
- On the other hand, that fat tail is the stickiest result of all my results. Color cuts on the background galaxies, aggressive masking, etc. do nothing. I had even hoped that CSFD, which is supposedly a superior extinction correction algorithm, would make it go away. No dice!
- Since what characterizes the redMaGiC sample is tightness in color space, I also tried to use the eBOSS LRG sample (maybe create one of my own?) but the source density was too low to be used.

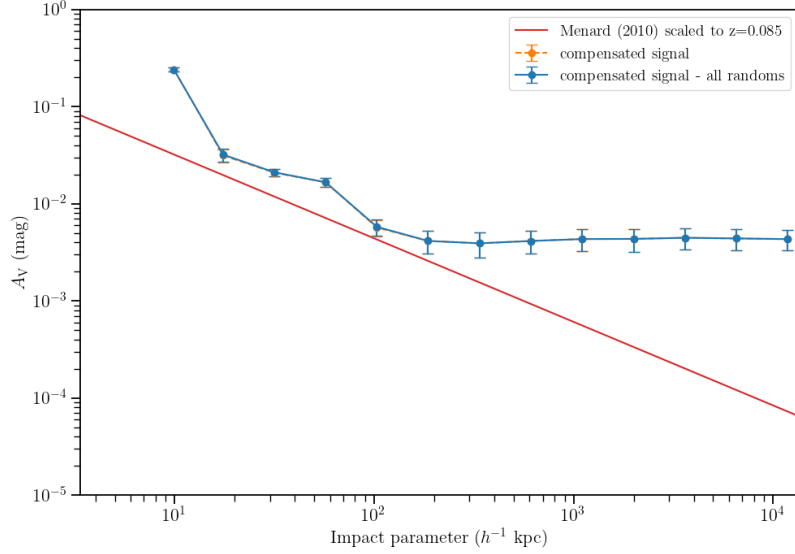


Figure 7: Extinction curve of WISExSuperCOSMOS foreground cross-correlated with SDSS background galaxies, shown as blue points. The M2010 prediction is shown in red.

5.3.2 GSWLC x SDSS

- This DOES look cool, particularly the fact that at higher impact parameters, the slope lines up very nicely with the redMaGiC samples.
- I did some testing with sSFR [show my algorithm?] but results were (at the time) mysterious to me. the selection functions are also quite heterogenous between quiescent and star-forming samples. Not sure how significant that latter fact is, but it's there.
- Important to mention: they all seem to depend heavily on the NSIDE of the mask I end up creating. This is a possible indication of my mask being incomplete or wrong; in any case this is why I don't 100% trust the result
- Plots to show: the neat spatial overlap plots, as well as the ugly multicolor plot which does a good job of demonstrating relationship between the various results, including this one.

Spatial overlap plot: put this into catalogs.

6 Discussion and Next Steps

6.1 Comment on results

- Fact of the matter is that I have been playing with this for years and the results have been consistent.
- The long time period has afforded me the opportunity to modernize, like 2023 dust relation, CSFD, WLS, etc.
- The fact that CSFD is high is consistent with the claim of [2023ApJ...958..118C] that “when SFD is used to derive absolute photometry for sources, Galactic extinction will be

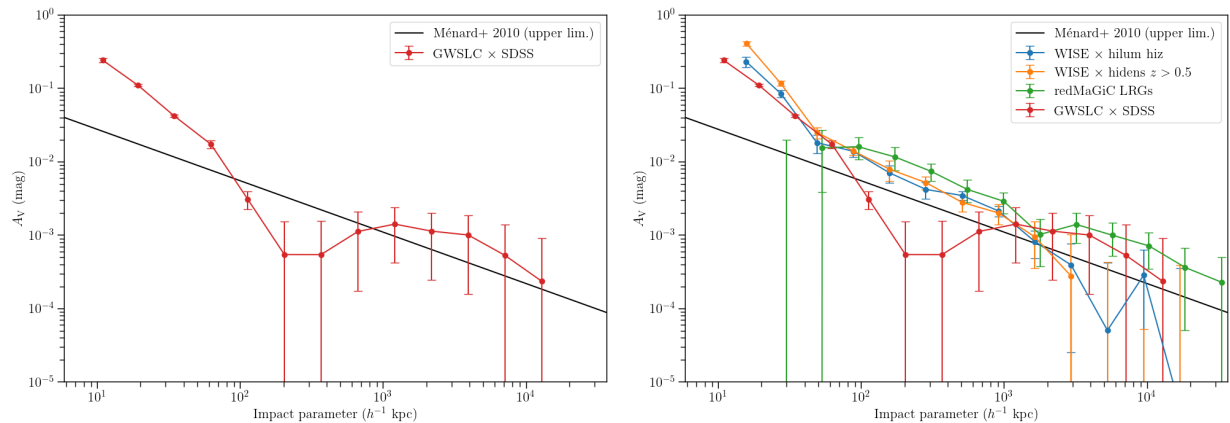


Figure 8: Extinction curve resulting from cross-correlation of GSWLC foreground and SDSS background galaxies. The GSWLC-SDSS result is shown in isolation in the left panel in red and together with the redMaGiC-based results in the right panel: WISExSuperCOSMOS–high-luminosity high-redshift (blue), WISExSuperCOSMOS–high density (orange), and the redMaGiC LRG fit (green). The M2010 prediction is shown in black in both panels.

overcorrected in a redshift- and spatially-dependent manner following the pattern of the cosmic web.” Now, which is more believable is at the reader’s discretion!

- Also, to some extent, it is to be expected that amplitude of signal is proportional to volume covered in foreground (right?). Then again, volume of Ménard sample is significantly higher than ours. As far as I know, that study has no selection for red sequence galaxies [Not sure what I meant by this – ed.]
- To end on a bright note, we know that the redMaGiC galaxies are biased tracers of LSS (by biased here I mean have a high galaxy bias relative to some field galaxy). The presence of dust well into the two-halo regime, while the WISExSuperCOSMOS samples (which presumably include star-forming galaxies) fall off, tells a cool story about redMaGiC as a tracer of LSS and WISExSuperCOSMOS a little more general.

6.2 Next steps

- At this point, I would like to publish. One can fiddle with results ad infinitum, but I’ve been urged by multiple people to publish so I will. If you all want to start running systematic analyses, you can, but I am not going to wait around.
- I figure this is a beginning, not an end. In follow-on studies, we can do calibration, segregate by galaxy type, etc. (Note that I had tried to do a galaxy type study (motivation for SDSS/GWSLC!) but my results were inconclusive and I haven’t redone them since.)
- For northern-sky samples, it might be worth trying Pan-STARRS instead of SDSS, or perhaps KiDS or some other such thing. I don’t know what the benefits would be of a more “modern” survey over SDSS, but it would be interesting to see.
- Jim, still waiting on your emission study! Also, Eric, could we transfer ownership of main repository to me? Alternatively, I can sync my fork.

- My ultimate goal is to make a function or tool that can help calibrate SN extinction along the line of sight as a function of impact parameter.

A Appendix: Weighted least-squares fit and results

The important parts of `do_WLS_fit.py` are reproduced below.

```
import statsmodels.api as sm
import numpy as np

def do_WLS_fit(this_dk, dust_rel, name=None, start=None, end=None, log=False):
    """
    Perform a weighted least-squares linear regression to
    data, taking care to transform the variables appropriately.

    Inputs
    this_dk: should be the 'compensated' treecorr output
    dust_rel: the corrected A_V relationship (dk-dr+rr)
    name: what are we calling it?
    start: exclude points with index < 'start' for fitting
    end: ibidem, but for points with index > 'end'
    log: if True, fit the relationship in log-space to stabilize the fits.
        Default is False.
    """

    ## If you want to exclude some points at the start/end
    dust_rel = dust_rel[start:end]
    this_dk = this_dk[start:end]

    ## If doing a logarithmic fit, transform the
    ## dependent variable (the fully compensated TreeCorr kappa)
    if log == True:
        # Keep positive values only
        ok = (dust_rel > 0)
        dust_rel = dust_rel[ok]
        this_dk = this_dk[ok]

        # Fitting log of dust relation, so take log
        Y = np.log(dust_rel)

        ## Also need to transform the weights!
        weights_OG = 1/(this_dk['sigma']**2)
        #weight_scale = (1/Y)**2
        weight_scale = (1/(Y+1))**2
        weights = weights_OG * weight_scale

    ## If not doing a logarithmic fit, don't transform
    else:
```

```

Y = dust_rel
weights = 1/(this_dk['sigma']**2)

## If plotting X on a log-scale, which we are,
## fit relationship on log scale
X = this_dk['meanlogr']

## Add intercept to abscissa
X = sm.add_constant(X)

## Do fit
mod_wls = sm.WLS(Y, X, weights=weights)
res_wls = mod_wls.fit()

## Print results
print("")
print(f"Fit result: {name}")
print("")
print(res_wls.summary())

return(res_wls)

```

Appendix: SDSS queries

Does this actually need to be here? Does anyone care?

B Full WLS Regression results

Full printouts of the `statsmodels.api.WLS` regressions are reproduced below. For real fits (not the systematics tests), the constant term is converted to a coefficient as e^{const} in the power law.

B.1 WISExSCOS and redMaGiC fits

The fit results below are based on only the “flat” portions of the extinction curves in Figure 3, i.e., 50-1000 kpc for the CSFD fits, and results are based on a high-redshift subset ($z_{\text{redmagic}} > 0.5$) of the redMaGiC high-density catalog.

WLS Regression Results			
=====			
Dep. Variable:	y	R-squared:	0.975
Model:	WLS	Adj. R-squared:	0.973
Method:	Least Squares	F-statistic:	397.0
Date:	Wed, 23 Oct 2024	Prob (F-statistic):	2.23e-09
Time:	18:59:41	Log-Likelihood:	8.2589
No. Observations:	12	AIC:	-12.52
Df Residuals:	10	BIC:	-11.55
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	-4.5308	0.068	-66.556	0.000	-4.683	-4.379
x1	-0.7677	0.039	-19.925	0.000	-0.853	-0.682
Omnibus:		1.628	Durbin-Watson:			2.041
Prob(Omnibus):		0.443	Jarque-Bera (JB):			1.154
Skew:		-0.692	Prob(JB):			0.562
Kurtosis:		2.373	Cond. No.			6.06

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

[-4.53082712 -0.7676558]

Fit: 1.08e-02 * (meanr)^{-0.77}

Below is the full fit result for the redMaGiC LRG sample, i.e., the low-redshift subset ($z_{\text{redmagic}} > 0.5$) of the redMaGiC high-density sample cross-correlated with the high-luminosity high-redshift

Fit result: redmagic LRG

WLS Regression Results

Dep. Variable:	y	R-squared:	0.943
Model:	WLS	Adj. R-squared:	0.936
Method:	Least Squares	F-statistic:	148.0
Date:	Wed, 23 Oct 2024	Prob (F-statistic):	6.85e-07
Time:	18:59:41	Log-Likelihood:	-2.4161
No. Observations:	11	AIC:	8.832
Df Residuals:	9	BIC:	9.628
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	-4.8020	0.225	-21.318	0.000	-5.312	-4.292
x1	-0.6872	0.056	-12.166	0.000	-0.815	-0.559
Omnibus:		7.605	Durbin-Watson:			1.874
Prob(Omnibus):		0.022	Jarque-Bera (JB):			3.122
Skew:		-1.072	Prob(JB):			0.210
Kurtosis:		4.487	Cond. No.			13.8

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

[-4.80200583 -0.68720111]

Fit: $8.21\text{e-}03 * (\text{meanr})^{-0.69}$

B.2 Gaia x redMaGiC

Gaia foreground cross-correlated with redMaGiC high-luminosity high-redshift sample.

Fit result: Gaia x hi-lum hi-z

WLS Regression Results						
=====						
Dep. Variable:	y	R-squared:	0.002			
Model:	WLS	Adj. R-squared:	-0.089			
Method:	Least Squares	F-statistic:	0.02226			
Date:	Wed, 23 Oct 2024	Prob (F-statistic):	0.884			
Time:	18:48:14	Log-Likelihood:	70.136			
No. Observations:	13	AIC:	-136.3			
Df Residuals:	11	BIC:	-135.1			
Df Model:	1					
Covariance Type:	nonrobust					
=====						
	coef	std err	t	P> t	[0.025	0.975]

const	-3.676e-05	0.000	-0.102	0.920	-0.001	0.001
x1	1.133e-05	7.59e-05	0.149	0.884	-0.000	0.000
=====						
Omnibus:	0.812	Durbin-Watson:	2.270			
Prob(Omnibus):	0.666	Jarque-Bera (JB):	0.628			
Skew:	0.015	Prob(JB):	0.730			
Kurtosis:	1.923	Cond. No.	51.2			
=====						

Gaia foreground cross-correlated with redMaGiC high-density sample.

Fit result: Gaia x high-dens

WLS Regression Results			
=====			
Dep. Variable:	y	R-squared:	0.049
Model:	WLS	Adj. R-squared:	-0.037
Method:	Least Squares	F-statistic:	0.5696
Date:	Wed, 23 Oct 2024	Prob (F-statistic):	0.466
Time:	18:48:14	Log-Likelihood:	69.030
No. Observations:	13	AIC:	-134.1
Df Residuals:	11	BIC:	-132.9
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	0.0003	0.000	0.740	0.475	-0.001	0.001
x1	-6.73e-05	8.92e-05	-0.755	0.466	-0.000	0.000
Omnibus:		0.378	Durbin-Watson:			3.063
Prob(Omnibus):		0.828	Jarque-Bera (JB):			0.490
Skew:		-0.175	Prob(JB):			0.783
Kurtosis:		2.115	Cond. No.			36.3

Finally, Gaia foreground cross-correlated with the combined redMaGiC galaxy sample.

Fit result: Gaia x combined redMaGiC

WLS Regression Results						
Dep. Variable:	y	R-squared:				0.006
Model:	WLS	Adj. R-squared:				-0.035
Method:	Least Squares	F-statistic:				0.1458
Date:	Wed, 02 Oct 2024	Prob (F-statistic):				0.706
Time:	18:24:35	Log-Likelihood:				138.85
No. Observations:	26	AIC:				-273.7
Df Residuals:	24	BIC:				-271.2
Df Model:	1					
Covariance Type:	nonrobust					
	coef	std err	t	P> t	[0.025	0.975]
const	0.0001	0.000	0.422	0.677	-0.000	0.001
x1	-2.139e-05	5.6e-05	-0.382	0.706	-0.000	9.42e-05
Omnibus:		1.496	Durbin-Watson:			2.023
Prob(Omnibus):		0.473	Jarque-Bera (JB):			0.966
Skew:		-0.059	Prob(JB):			0.617
Kurtosis:		2.063	Cond. No.			45.5

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
[1.11602366e-04 -2.13928514e-05]

Gaia x combined redMaGiC WLS fit:

Const: 1.116e-04 +/- 2.646e-04
Slope: -2.139e-05 +/- 5.603e-05

B.3 WISExSuperCOSMOS x DES stars

Standard WISExSuperCOSMOS foreground sample cross-correlated with DES Y3 star sample background. The rand2 refers to the new random-catalog generation algorithm I used, which

Fit result: WISExSCOS rand2 x DES star

WLS Regression Results						
=====						
Dep. Variable:	y	R-squared:	0.147			
Model:	WLS	Adj. R-squared:	0.070			
Method:	Least Squares	F-statistic:	1.900			
Date:	Wed, 02 Oct 2024	Prob (F-statistic):	0.195			
Time:	18:24:36	Log-Likelihood:	69.816			
No. Observations:	13	AIC:	-135.6			
Df Residuals:	11	BIC:	-134.5			
Df Model:	1					
Covariance Type:	nonrobust					
=====						
	coef	std err	t	P> t	[0.025	0.975]

const	-0.0005	0.000	-1.188	0.260	-0.002	0.000
x1	0.0002	0.000	1.378	0.195	-0.000	0.000
=====						
Omnibus:	0.407	Durbin-Watson:	1.947			
Prob(Omnibus):	0.816	Jarque-Bera (JB):	0.509			
Skew:	-0.268	Prob(JB):	0.775			
Kurtosis:	2.193	Cond. No.	7.93			
=====						

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
[-0.00053476 0.000177]

WISExSCOS updated rand x DES star WLS fit results:

Coeff: -5.348e-04 +/- 4.500e-04

Slope: 1.770e-04 +/- 1.284e-04

B.4 GWSLC x SDSS

This is the fit result for a subset of the GWSLC X-2 foreground catalog cross-correlated with a background catalog selected from SDSS. In this case,

Fit result: partial GWSLC x SDSS

WLS Regression Results			
=====			
Dep. Variable:	y	R-squared:	0.969
Model:	WLS	Adj. R-squared:	0.961

```

Method:                Least Squares    F-statistic:                124.3
Date:                  Tue, 15 Oct 2024  Prob (F-statistic):        0.000368
Time:                  09:08:29          Log-Likelihood:            0.91149
No. Observations:      6                AIC:                      2.177
Df Residuals:          4                BIC:                      1.761
Df Model:              1
Covariance Type:       nonrobust

```

	coef	std err	t	P> t	[0.025	0.975]
-----	-----	-----	-----	-----	-----	-----
const	-4.2028	0.264	-15.926	0.000	-4.935	-3.470
x1	-0.7612	0.068	-11.148	0.000	-0.951	-0.572
=====	=====	=====	=====	=====	=====	=====
Omnibus:		nan	Durbin-Watson:			1.677
Prob(Omnibus):		nan	Jarque-Bera (JB):			0.340
Skew:		0.319	Prob(JB):			0.844
Kurtosis:		2.025	Cond. No.			10.5
=====	=====	=====	=====	=====	=====	=====

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
 [-4.20276501 -0.7611631]

GWSLC x SDSS Fit: 1.45e-02 * (meanr)^{-0.75}
 partial GWSLC x SDSS Fit: 1.50e-02 * (meanr)^{-0.76}

References

- [1] Karl D. Gordon. “dust_extinction: Interstellar Dust Extinction Models”. In: *Journal of Open Source Software* 9.100 (2024), p. 7023. DOI: 10.21105/joss.07023. URL: <https://doi.org/10.21105/joss.07023>.
- [2] Karl D. Gordon et al. “One Relation for All Wavelengths: The Far-ultraviolet to Mid-infrared Milky Way Spectroscopic R(V)-dependent Dust Extinction Relationship”. In: *Astrophysical Journal* 950.2, 86 (June 2023), p. 86. DOI: 10.3847/1538-4357/accb59. arXiv: 2304.01991 [astro-ph.GA].